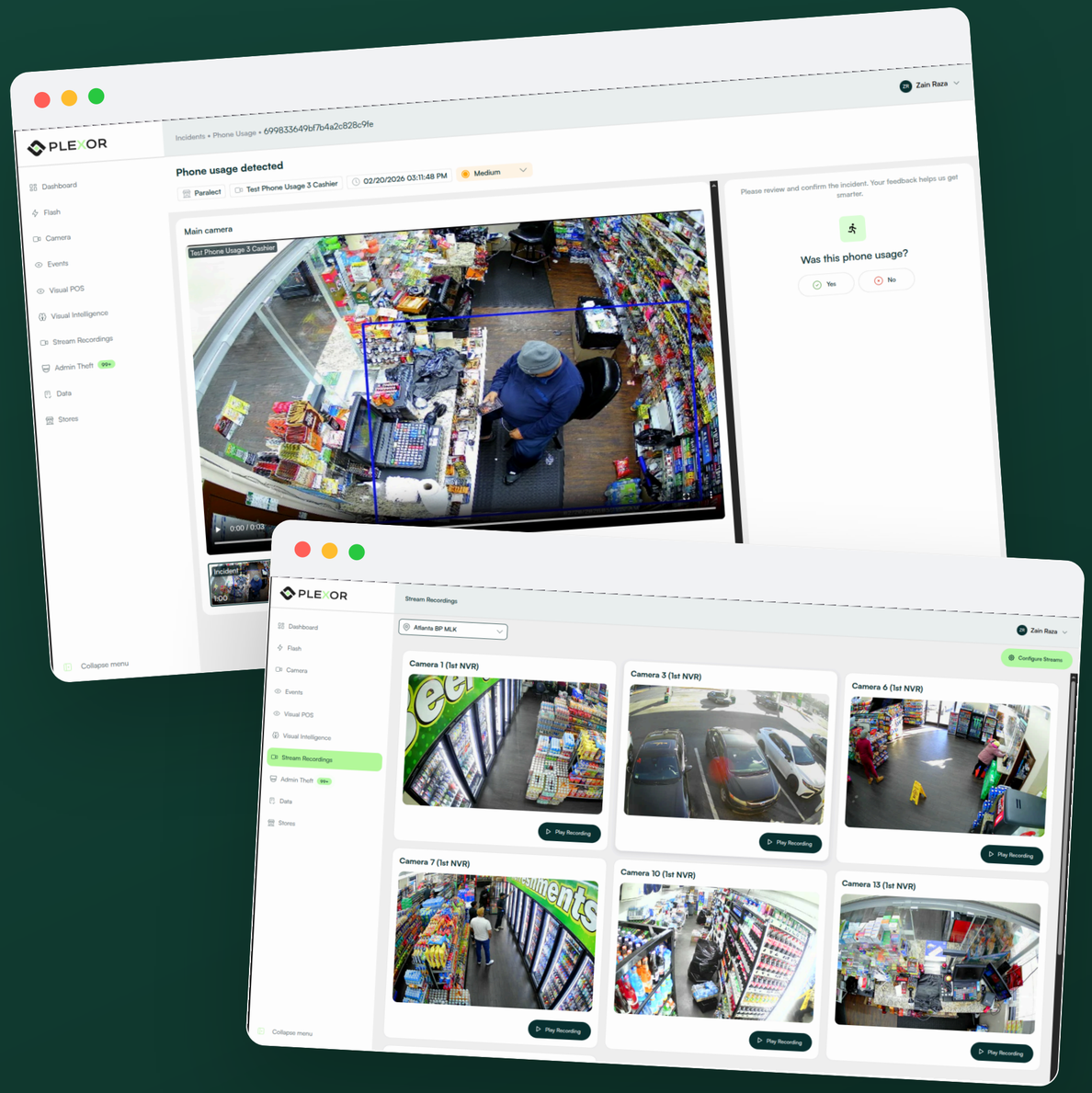
PLEXOR

05 / AI SURVEILLANCE

AI-Based Surveillance System

Computer-vision loss-prevention for retail — theft, POS fraud and loitering, flagged in real time.



Zohaib Ali Mughal · Full-stack engineer

PLEXOR

05 — OVERVIEW

AI-Based Surveillance System

Retail shrinkage from employee & customer theft and POS under-ringing is hard to catch live. A multi-camera computer-vision platform detects it and routes alerts to store and district managers.

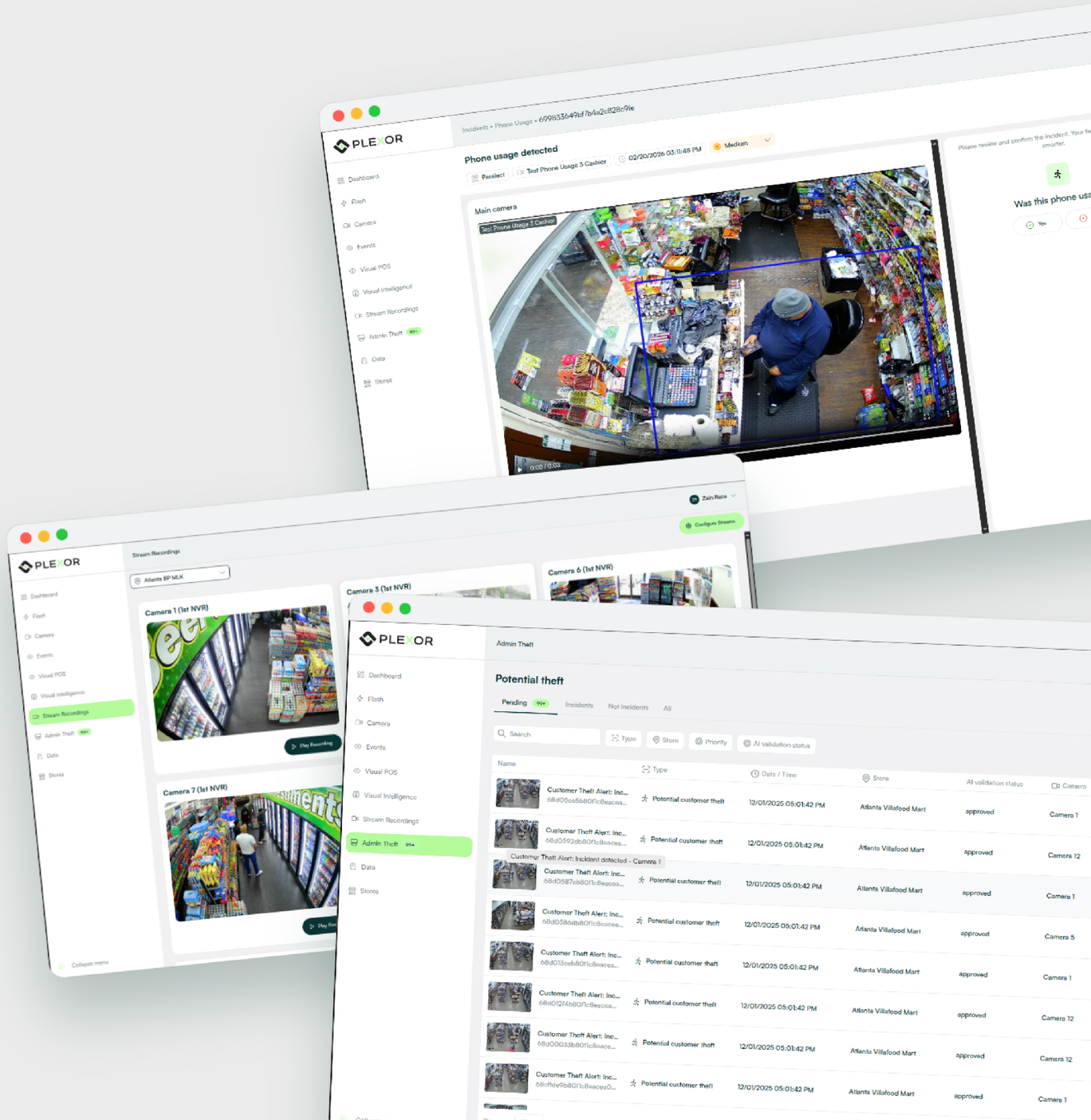
ROLE **Full-stack engineer**

100+
STORES MONITORED

~700ms
REALTIME INFERENCE

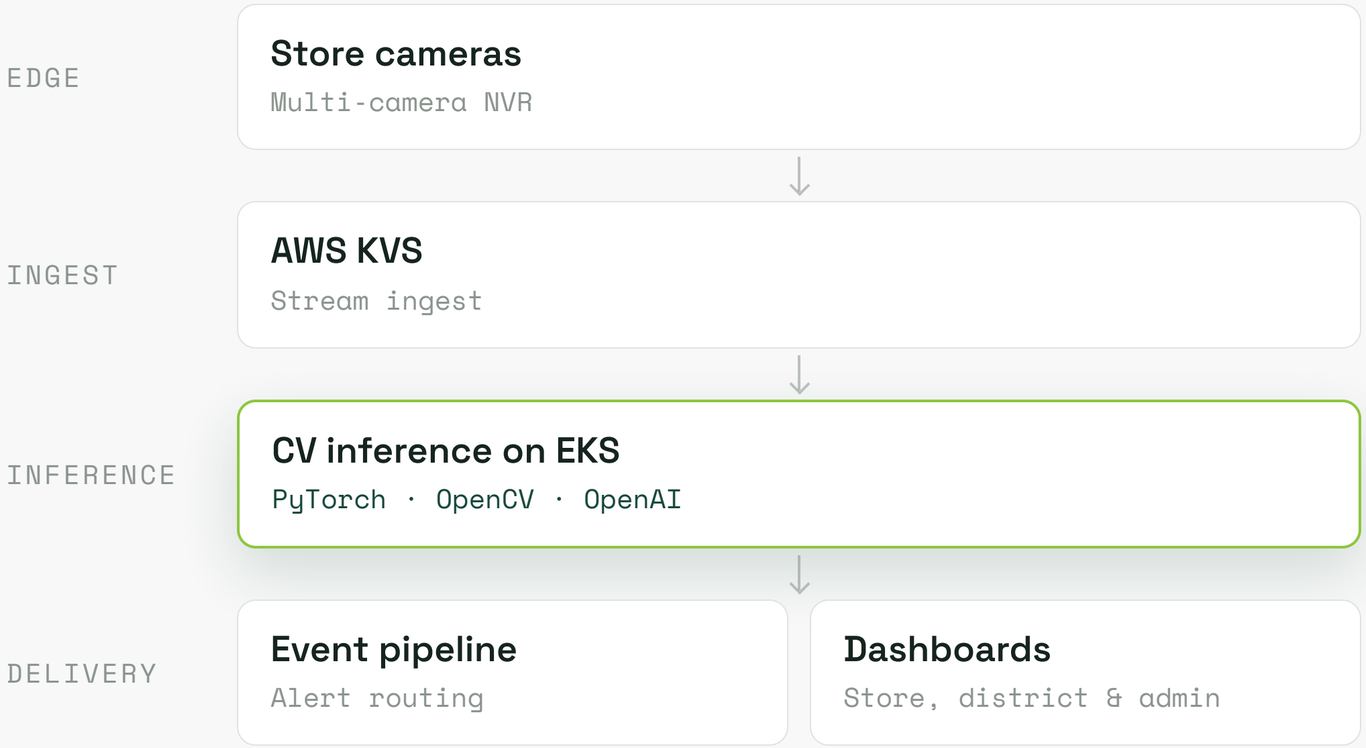
37+
DETECTION MODELS

- Theft detection across employees & customers
- Phone-usage & loitering alerts with live recording streams
- Visual POS & under-ringing detection



05 — ARCHITECTURE & STACK

Realtime vision pipeline, role-scoped delivery



KEY DECISIONS

01 Realtime multi-camera streaming over AWS KVS for low-latency feeds

02 Model serving on EKS for scalable GPU inference

03 Role-scoped dashboards for store, district and admin views

STACK

FRONTEND: Next.js, Mantine

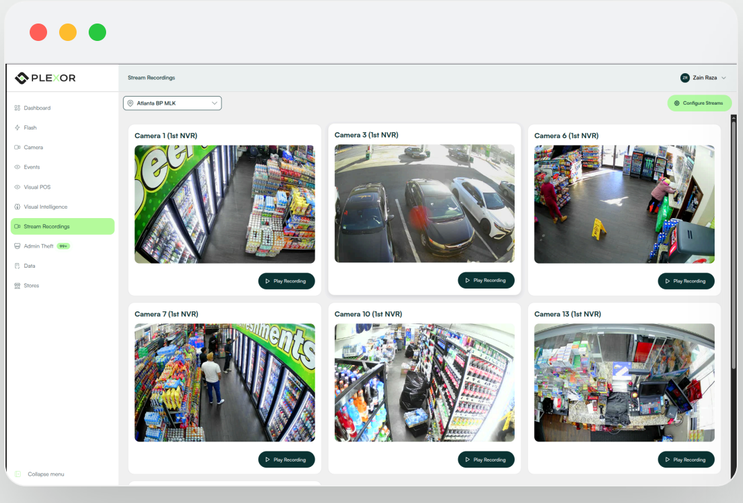
BACKEND: Node.js, Fastify, Python

CLOUD: AWS, EKS, Lambda, S3, KVS

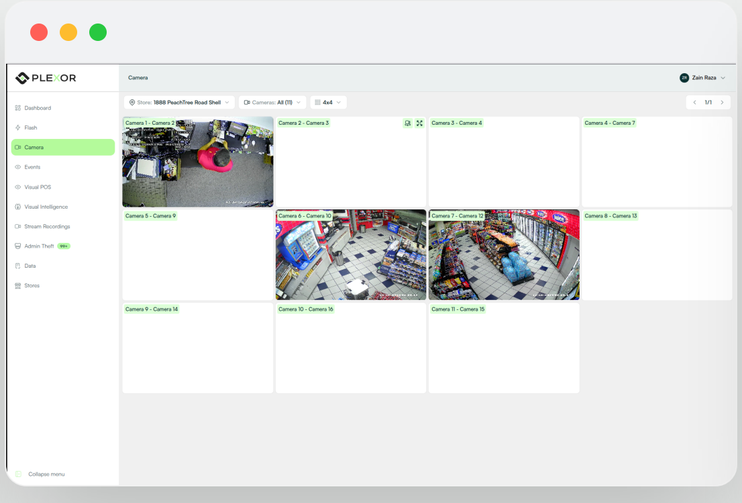
AI: PyTorch, OpenCV, OpenAI

Inside the product

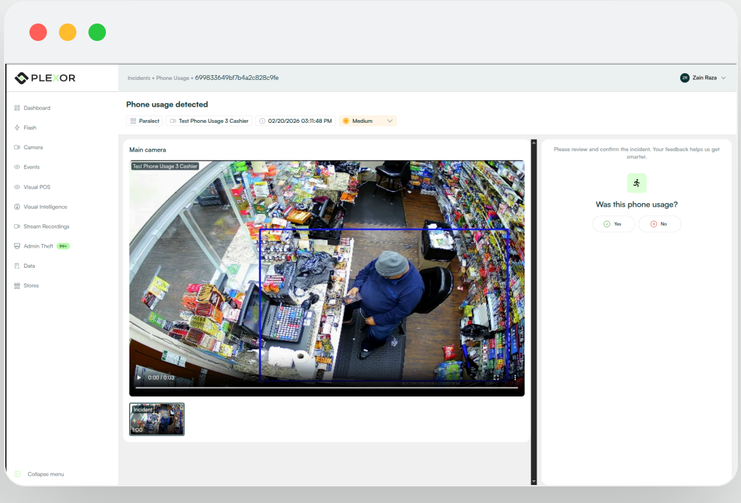
PLEXOR



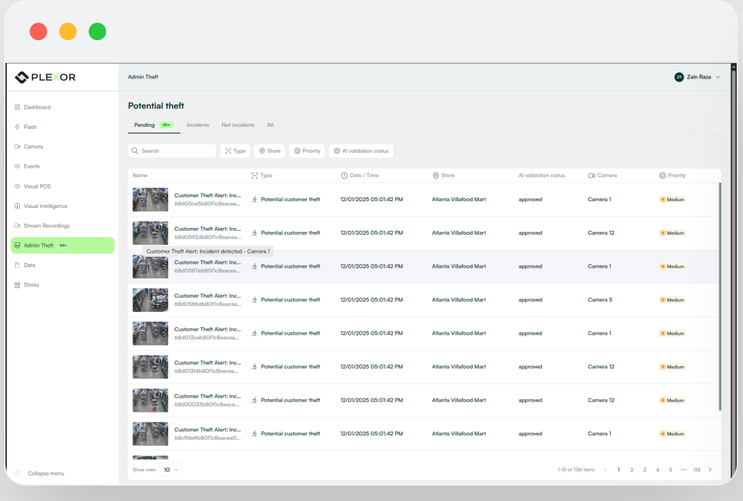
Live recording streams



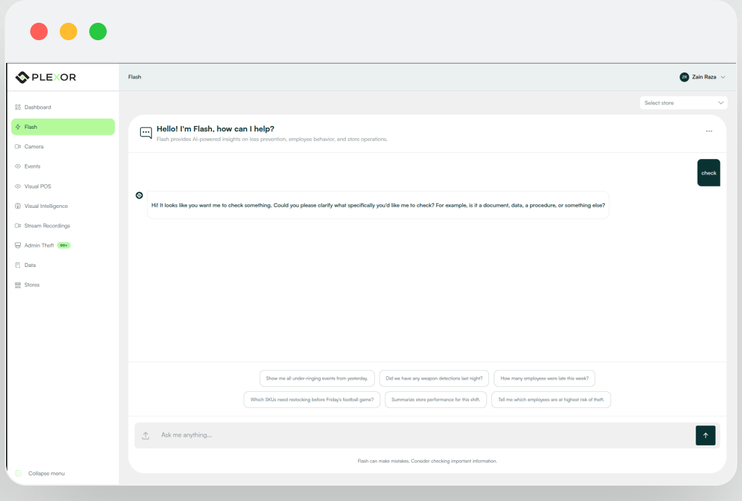
Multi-camera live grid



Phone-usage detection & review



Theft queue & AI validation



Flash: AI loss-prevention assistant

WHAT IT DOES

Watches every store camera, flags theft, phone use, loitering and POS under-ringing, and routes prioritised alerts to store and district managers.

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05 — IMPACT

Realtime detection at chain scale

100+
STORES MONITORED

~700ms
REALTIME INFERENCE

37+
DETECTION MODELS

Surfaced theft and POS under-ringing events in real time across monitored stores, routing prioritised alerts to the managers who could act on them.

WHAT SHIPPED

- Theft detection across employees and customers
- Phone-usage and loitering alerts with live streams
- Visual POS and under-ringing flags
- Prioritised alerts to store and district managers

Zohaib Ali Mughal z.mughal.24648@gmail.com zohaibmughal.dev/work/plexor